

a) Best Fit

```
#include <stdio.h>

int main() {
    int m, n;

    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);

    int block[m], remainingSize[m];

    printf("Enter size of each block:\n");
    for(int i = 0; i < m; i++) {
        scanf("%d", &block[i]);
        remainingSize[i] = block[i];    // initialize remaining memory
    }

    printf("Enter number of processes:\n");
    scanf("%d", &n);

    int process[n];

    printf("Enter size of each process:\n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &process[i]);
    }

    // Best Fit Logic with remaining memory
    for(int i = 0; i < n; i++) {
        int best = -1;

        for(int j = 0; j < m; j++) {
            if(remainingSize[j] >= process[i]) {
                if(best == -1 || remainingSize[j] < remainingSize[best]) {
                    best = j;
                }
            }
        }

        if(best != -1) {
            int fragment = remainingSize[best] - process[i];

            printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n",
                i + 1, process[i], best + 1, block[best], fragment);

            remainingSize[best] -= process[i];    // update remaining memory
        }
        else {
            printf("Process %d of size %d is not allocated\n",
                i + 1, process[i]);
        }
    }

    return 0;
}
```

Custom Test

Success

Input :

```
5
100 500 200 300 600
4
212 417 112 426
```

Expected Output :

```
Enter number of memory blocks:
Enter size of each block:
Enter number of processes:
Enter size of each process:
Process 1 of size 212 is allocated to Block 4 of size 300 with
Fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500 with
Fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200 with
Fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600 with
Fragment 174
```

Output :

```
Enter number of memory blocks:
Enter size of each block:
Enter number of processes:
Enter size of each process:
Process 1 of size 212 is allocated to Block 4 of size 300 with Fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500 with Fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200 with Fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600 with Fragment 174
```

b) Fitst Fit

```
#include <stdio.h>

int main() {
    int m, n;

    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);

    int blockSize[m], remainingSize[m];

    printf("Enter size of each block:\n");
    for(int i = 0; i < m; i++) {
        scanf("%d", &blockSize[i]);
        remainingSize[i] = blockSize[i]; // fixed typo
    }

    printf("Enter number of processes:\n");
    scanf("%d", &n); // fixed syntax

    int processSize[n];

    printf("Enter size of each process:\n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
    }

    // First Fit Allocation
    for(int i = 0; i < n; i++) {
        int allocated = 0;

        for(int j = 0; j < m; j++) {
            if(remainingSize[j] >= processSize[i]) {
                int fragment = remainingSize[j] - processSize[i];

                printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n",
                    i + 1, processSize[i], j + 1, blockSize[j], fragment);

                remainingSize[j] -= processSize[i];
                allocated = 1;
                break;
            }
        }

        if(!allocated) {
            printf("Process %d of size %d is not allocated\n",
                i + 1, processSize[i]);
        }
    }

    return 0;
}
```

✓ Custom Test

Success ☺

Input :

```
5
100 500 200 300 600
4
212 417 112 426
```

Expected Output :

```
Enter number of memory blocks:
Enter size of each block:
Enter number of processes:
Enter size of each process:
Process 1 of size 212 is allocated to Block 2 of size 500 with
Fragment 288
Process 2 of size 417 is allocated to Block 5 of size 600 with
Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 500 with
Fragment 176
Process 4 of size 426 is not allocated
```

Output :

```
Enter number of memory blocks:
Enter size of each block:
Enter number of processes:
Enter size of each process:
Process 1 of size 212 is allocated to Block 2 of size 500 with Fragment 288
Process 2 of size 417 is allocated to Block 5 of size 600 with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 500 with Fragment 176
Process 4 of size 426 is not allocated
```